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Scalzo et al.

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(54) **BLUEBERRY PLANT NAMED 'C15-143'**

(50) Latin Name: ***Vaccinium corymbosum* hybrid**
Varietal Denomination: **C15-143**

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A01H 5/08 (2018.01)
A01H 6/36 (2018.01)

(52) **U.S. Cl.**
USPC **Plt./157**
CPC ***A01H 6/368*** (2018.05)

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CPC ... A01H 5/08; A01H 5/00; A01H 5/02; A01H 6/36; A01H 6/368
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

PP19,503 P3 * 11/2008 Lyrene A01H 6/368 Plt./157
PP20,695 P2 * 2/2010 Wright Plt./157

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(57) **ABSTRACT**

The new blueberry plant variety 'C15-143' is provided. 'C15-143' is a commercial variety intended for the fresh market. The variety is produced from a cross of unpatented parents 'FL12-082' and 'FL12-069'.

5 Drawing Sheets

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Latin name of the genus and species:
Genus—*Vaccinium.*
Species—*corymbosum* hybrid.
Variety denomination: The new blueberry plant claimed is of the variety denominated 'C15-143'.

BACKGROUND OF THE INVENTION

The new variety 'C15-143' was selected from a population of seedlings derived from crossing the blueberry varieties known as 'FL12-082' (unpatented seed parent) and the variety known as 'FL12-069' (unpatented pollen parent). The cross was made in 2012 in Florida, USA and the seed was sown and grown in Corindi Beach, New South Wales, Australia. The new variety was selected in 2015 from among plants located on land at Corindi Beach and assigned the breeding code 'C15-143'. Plants of 'C15-143' were propagated by cuttings for further evaluation and resulted to be uniform and stable. The new variety showed distinctive traits such as evergreen, medium plant vigorour, early season crop, and firm fruit.

SUMMARY OF THE INVENTION

Asexual reproduction of the new variety 'C15-143' by cutting propagation since 2015 at Corindi Beach, New South Wales, Australia has demonstrated that the new variety reproduces true to type plants.

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The new variety was selected in 2015 as a single plant within a population of seedlings resulting from controlled cross of *Vaccinium* varieties. The seedling population was planted in an experimental block in the field at Corindi Beach, New South Wales, Australia and the selection of the new variety took place in the same block. Selection criteria were a combination of early season cropping time, low chilling requirement, medium plant vigour, non-deciduous type of plant (evergreen), high yields, and firm fruit. The new variety was subsequently evaluated for five years at the commercial farm at Corindi Beach, New South Wales, Australia.

The following characteristics of the new variety have been repeatedly observed and can be used to distinguish 'C15-143' as a new and distinct variety of *Vaccinium corymbosum* hybrid:

1. Non-deciduous (Evergreen)
2. Early season crop
3. Medium to high yield
4. Low chilling requirement, estimated to be between 150 and 300 hours
5. Firm fruit

The new blueberry variety 'C15-143' has maintained its distinguished characteristics throughout successive asexual propagation. The variety has been repeatedly asexually reproduced through softwood cuttings in New South Wales, Australia and the clones are phenotypically identical to the original plant.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying photographic illustration shows typical specimens in full color of the foliage and fruit of the new variety 'C15-143'. The colors are as nearly true as is reasonably possible in a color representation of this type.

FIG. **1** is a photograph of the new variety 'C15-143', showing three year old plants growing in pots.

FIG. **2** is a photograph of the new variety 'C15-143', showing the botanical parts of the plants (blue fruit, green fruit, flowers and leaves).

FIG. **3** is a photograph of the typical fruit cluster of the variety 'C15-143'.

FIG. **4** is a photograph showing the typical fruit calyx of the variety 'C15-143'.

FIG. **5** is a photograph showing the typical pedicel scar of the fruit of the variety 'C15-143'.

The colors in the photographs are as close as possible with the photographic and printing technology utilized. The color values cited in the detailed botanical description accurately describe the colors of the new blueberry variety.

DETAILED BOTANICAL DESCRIPTION

The following detailed description sets forth the distinctive characteristics of 'C15-143'. The data which defines these characteristics was collected from asexual reproductions of the original selection. Dimensions, sizes, colors, and other characteristics are approximations and averages set forth as accurately as possible. For all traits data was collected from 3 plant parts across 6 randomly selected plants. For the traits relating to fruits (e.g., fruit weight, firmness, brix, acidity) the data is an average across twenty fruits collected randomly. The plant history was taken on plants approximately 3 years of age, and the descriptions relate to plants grown in the field in Corindi Beach, New South Wales, 2456 Australia. Descriptions of fruit characteristics were made on fruit grown in Corindi Beach, New South Wales, 2456 Australia. Color designations are from the 2007 edition of The Royal Horticultural Society ("R.H.S.") Colour Chart. 'C12-122' has not been observed under all possible environments. The phenotype may vary slightly with different growing environments such as temperature, light, fertility, soil pH, moisture and maturity levels, but without any change in the genotype.

Classification:
- a. *Family*.—Ericaceae.
- b. *Genus*.—*Vaccinium*.
- c. *Species*.—*Corymbosum* hybrid.
- d. *Common name*.—Blueberry.

Parentage:
- *Female parent*.—Unpatented selection 'FL12-082'.
- *Male parent*.—Unpatented selection 'FL12-069'.

Market class: Fresh market.

PLANT

General:
- *Parentage*.—FL12-082 x FL12-069.
- *Plant height*.—1.53 m.
- *Plant width*.—1.2 m.
- *Growth habit*.—Semi-upright.
- *Growth*.—Medium vigour.
- *Productivity*.—Medium to high yield. Average of 4.7 Kg per plant (estimated equivalent production of 10.3 pounds per plant) from 3 year old plants when growing in 17 L pots at Corindi Beach, NSW. The plants are spaced at 0.7 m apart along the row and 2.5 m between the rows, which gives an estimated plant density of 5700 plants per hectare.
- *Mature cane color*.—Grey-orange group 165A, and streak of grey-brown group, similar to 199C.
- *One-year-old shoot's color designation*.—The base color is green group, similar to 143A.
- *One-year-old shoot's average length of the internode*.—34.4 mm.
- *Cold hardiness*.—Low chill (USDA plant hardiness Zone 4).
- *Cold tolerance*.—Low.
- *Chilling requirement*.—Low, estimated between 150 and 300 hours.
- *Tolerance to disease*.—None assessed.
- *Leafing*.—Overall medium, plant retains leaves during the winter in low chill environments (zone 4).
- *Twigginess*.—Medium.

FOLIAGE

General:
- *Leaf color* (*top side*).—Green group, similar to 137A.
- *Leaf color* (*under side*).—Green group, similar to N138C.
- *Leaf arrangement*.—Alternate.
- *Leaf shape*.—Elliptic.
- *Leaf margins*.—Minor serration
- *Leaf venation*.—Reticulate.
- *Leaf length*.—Very short to short (average 47.5 mm).
- *Leaf width*.—Very narrow to narrow (average 23.9 mm).
- *Leaf length/width ratio*.—2.
- *Shape of the leaf apex*.—Acute.
- *Shape of the leaf base*.—Cuneate.
- *Leaf nectaries*.—Absent.
- *Pubescence of upper side*.—Absent.
- *Pubescence of lower side*.—Absent.
- *Cross sectional profile*.—Flat.
- *Longitudinal profile*.—Straight.
- *Attitude*.—Horizontal.

Petioles:
- *Length*.—Average 4.2 mm.
- *Width*.—Average **1**.26 mm.
- *Color*.—Yellow-green group, similar to 145B.
- *Texture*.—Medium to rough.

FLOWERS

General:
- *Time of beginning of flowering*.—Early season (50% of anthesis estimated to be on the 1st of June, on 3 year old plants, cultivated at Corindi Beach, NSW.
- *Flower shape*.—Urceolate.
- *Flower fragrance*.—Yes, perceptible.

Corolla:
- *Shape*.—Urceolate.
- *Color*.—White group similar to NN155B.
- *Length*.—7.45 mm.
- *Width*.—7.2 mm.
- *Aperture width*.—3.38 mm.
- *Anthocyanin coloration of corolla*.—Absent or very weak.
- *Corolla ridges*.—Present, average of 5.
- *Petal width* (*ridge to ridge*).—4.89 mm.

Protrusion of stigma.—Very small present.

Corolla/petal texture.—Smooth.

Calyx (with sepals):

Diameter.—5.99 mm.

Color of sepal's outside.—Green group, similar to 138B.

Color of sepal's inside.—Green group, similar to 138A.

Inflorescence:

Inflorescence length (excluding peduncle).—Long, 21.5 mm.

Inflorescence width.—19.10 mm.

Surface texture of peduncle.—Medium (texture between rough and smooth).

Color of peduncle.—Base color is yellow-green group 145A, over color is grey-red group, similar to 182A.

Length of pedicel.—10.3 mm.

Surface texture of pedicel.—Smooth.

Color of pedicel.—Yellow-green group, similar to 145A and 146B.

Number of flowers per cluster.—6.

Flower cluster density.—Medium.

Flowering interval on one-year old shoot.—June to end of July.

Flowering interval on current year shoot.—June to end of July.

Stamen:

Length.—6.43 mm.

Number per flower.—10.

Filament color.—Yellow-green group, similar to 145D.

Pistil:

Pistil length (including ovary).—10.2 mm.

Style length (including stigma).—8.7 mm.

Style color.—Yellow-green group, similar to N144B.

Anther:

Length.—3.8 mm.

Number.—10.

Color.—Greyed-orange group, similar to 165A and 165B.

Pollen:

Self compatibility.—Yes, this variety shows a high degree of self-compatibility.

Abundance.—Low.

Color.—Yellow group 11C.

FRUIT

General:

Time of fruit ripening.—Early, estimated 50% of the fruit ripe on the 15^{th} of August, on 3 year old plants, growing at Corindi Beach, NSW.

Plant fruiting type.—One-year-old and current seasons shoots.

Cluster density.—Medium, average 6 berries per cluster.

Unripe fruit color.—Yellow-green group, similar to 144B.

Ripe berry color.—Blue group, similar to 103A, when bloom is removed.

Berry surface wax abundance.—Strong.

Berry weight.—On average 2.6 g.

Berry height from calyx to scar.—13.6 mm.

Berry diameter.—15.3 mm.

Berry shape.—Oblate.

Fruit diameter of calyx basin.—Small to medium, on average 5.4 mm.

Fruit depth of calyx basin.—Deep.

Fruit stem scar.—Medium size scar.

Sweetness when ripe.—High (14 Brix).

Firmness when ripe.—Very firm, 220 g/mm, measured with FirmTech.

Acidity when ripe.—Medium (0.6%).

Berry flesh color.—Yellow-green group, similar to 145B and 145C.

Storage quality.—Medium average 22 days.

Suitability for mechanical harvesting.—Not tested.

Self-fruitfulness.—Yes.

Uses.—Fruit to be hand harvested for fresh market.

SEED

General:

Seed abundance in fruit.—Low, 8 seeds per fruit.

Seed color.—Greyed orange group, similar to 165A.

Seed length.—1.9 mm.

COMPARISON WITH SIMILAR CULTIVARS

Table 1 below provides a comparison between 'C15-143' and several similar cultivars:

TABLE 1

Comparison of 'C15-143' with similar cultivars

Characteristics	'C15-143'	'C99-42' (U.S. Plant Pat. No. 20695P2)	'Snowchaser' (U.S. Plant Pat. No. 19503P3)
Plant vigour	Medium	Weak to medium	Medium
Plant growth habit	Semi-upright	Semi-upright to intermediate	Semi-upright
One year old shoot length of internodes	Short	Very short to short	Short
Leaf length	Very short to short	Short	Long
Leaf length (mm)	47.5 ± 0.6	53.10 ± 0.8	61.50 ± 0.9
Leaf width	Very narrow to narrow	Very narrow to narrow	Broad
Leaf width (mm)	23.9 ± 1.3	22.9 ± 1.3	34.3 ± 1.5
Flower size of corolla	Small	Medium	Medium
Flower corolla length	7.4 ± 0.3	9.7 ± 0.3	9.5 ± 0.3
Fruit cluster density	Medium	Sparse	Medium
Fruit size	Medium to large	Small to medium	Small to medium
Fruit weight (g)	2.6 ± 0.3	1.9 ± 0.39	1.7 ± 0.21
Fruit diameter (mm)	15.3 ± 0.7	15.6 ± 0.96	15.10 ± 0.97
Fruit depth of calyx basin	Deep	Medium	Shallow
Fruit intensity of bloom	Strong	Weak to medium	Weak to medium
Fruit firmness	Firm	Firm	Soft
Soluble solid content (%)	14	13.1	14.3
Titratable acidity (%)	0.6	0.3	0.5
Time of vegetative bud burst	Medium	Early	Early
Time of beginning of flowering	Early	Early to medium	Very early to early
Time of beginning of fruit ripening	Early	Early to medium	Very early to early

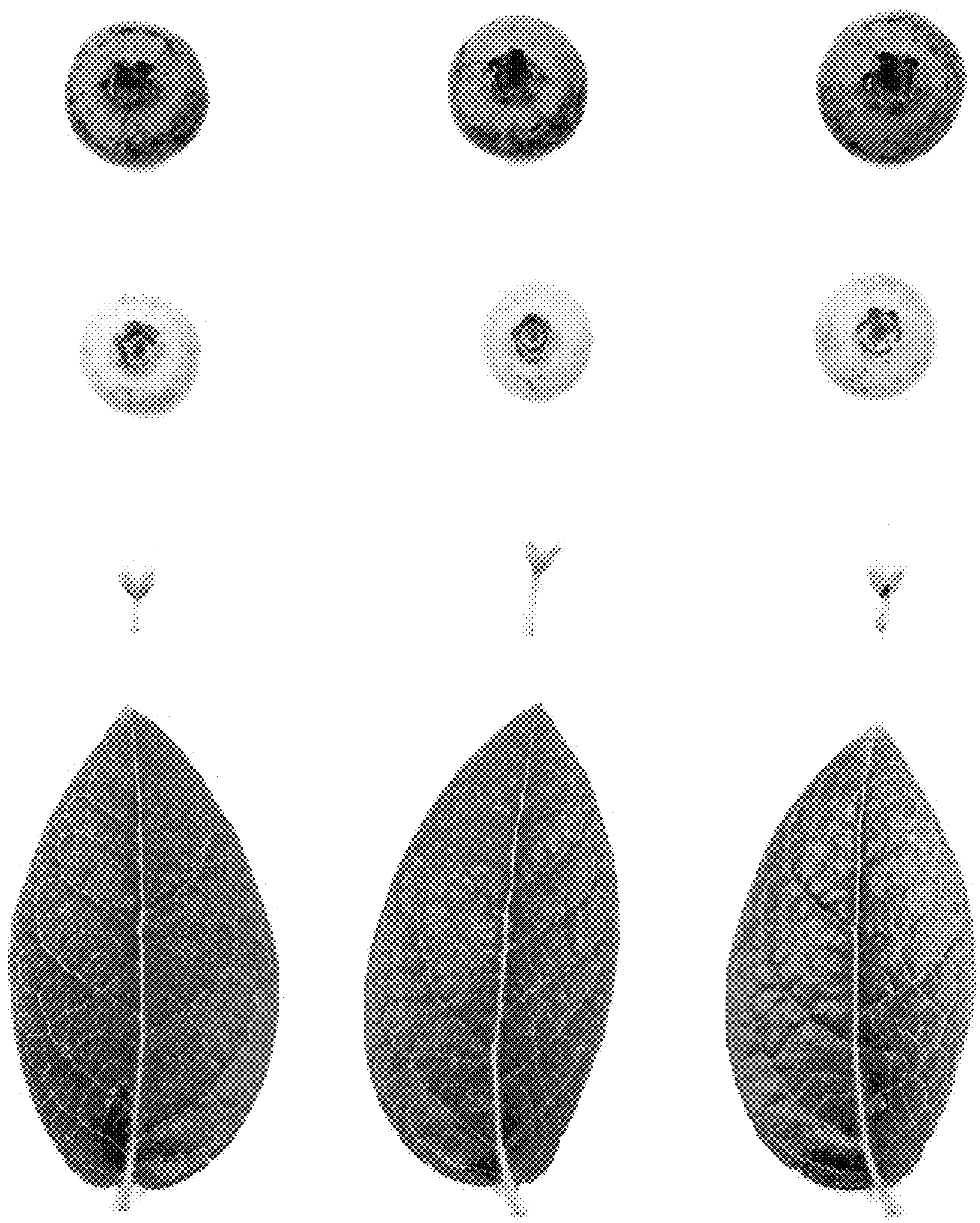
The invention claimed is:

1. A new and distinct variety of blueberry plant named 'C15-143', substantially as illustrated and described herein.

* * * * *

FIG. 1

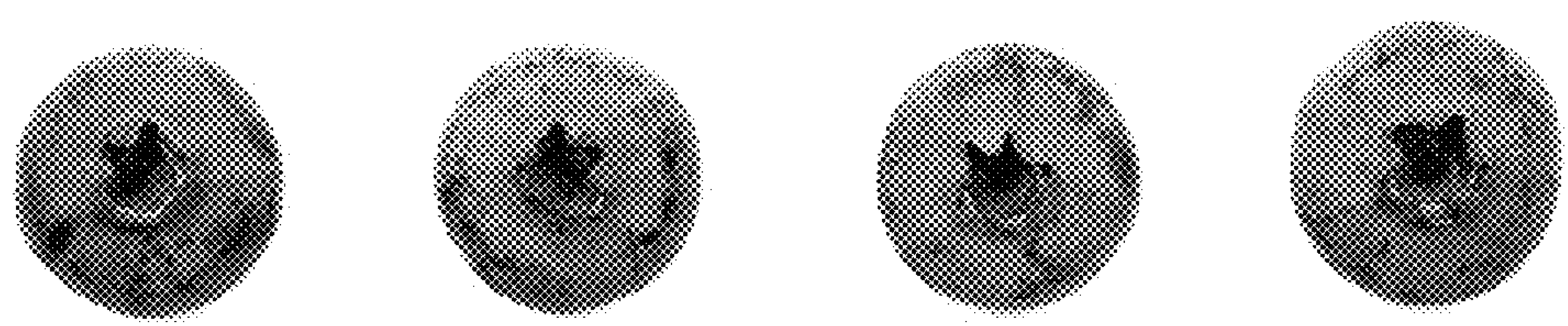
FIG. 2



C15-143

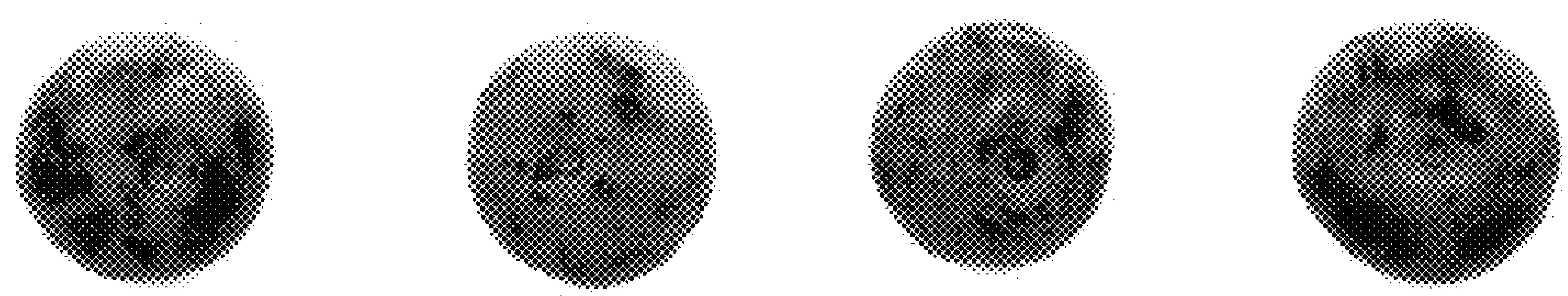
FIG. 3

FIG. 4



C15-143

FIG. 5



C15-143